

Paul Abrams

Toronto, Ontario

Senior Software Developer · .NET Core / C# · Node.js / NestJS · Cloud-Native Microservices & REST APIs · Docker / Kubernetes · Azure & CI/CD · 15+ Years High-Availability Distributed Government Systems

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Professional Summary

Senior software developer with 15+ years building, scaling, and operating high-availability, distributed enterprise systems across the Government of Canada — from a **load-balanced, geographically distributed** legal case-management platform serving **5,000+ concurrent users nationally** to a **cloud-native microservices** application on Microsoft Azure. Deep **backend** expertise in **.NET Core / C#** and **Node.js / NestJS** (a TypeScript decorator framework analogous to **Spring Boot / ASP.NET Core**), designing and consuming **REST APIs between services** and publishing/governing them through an **API gateway (Azure API Management)** with authentication, throttling, and transformation. Hands-on with **Docker and Kubernetes** containerized deployment, **CI/CD pipelines** (Azure DevOps, GitHub Actions), **serverless / event-driven workflows** (Azure Functions, queue- and storage-triggered handoffs), and **load balancing** across cloud and on-prem tiers. Strong relational-database engineering (**MS SQL Server, Azure SQL, PostgreSQL**) with working **NoSQL (MongoDB)** exposure, and a track record of **optimizing high-traffic backends** (e.g. a 50× report speedup via execution-plan analysis). **Security-minded by default:** OAuth 2.0 / OpenID Connect / Entra IAM, migration to **Managed Identities, Key Vault** secret hygiene, TLS / certificate management, and **ITSG-33 / PBMM** controls. **Recurring pattern:** self-initiated automation frameworks that pay back substantial labour cost. **Professional Scrum Master I; AODA / WCAG-aware.** Active open-source contributor — **60+ public PRs** including Microsoft's `vsts-team-calendar`, `tfs-cli`, and the **Microsoft Authentication Library .NET** docs.

Professional Experience

Abrams Software Kingston / Toronto, Ontario (Remote)

August 2025 – Present

Full-Stack Developer · Independent

- **TWNR** — architected and shipped a public-facing multiplayer game as a full-stack web application, solo. Dockerized **TypeScript** monorepo: **Node.js (Express) + WebSocket** backend backed by **PostgreSQL**, per-player **JWT auth**, and a **JSON-Schema-generated REST API contract** that doubles as live documentation. Concurrency-correct state via PostgreSQL transactions (`SELECT ... FOR UPDATE`). Live at twnr.paulabrams.ca.
- **Chess Attack Visualization** — **React 19 + TypeScript** app consuming the **Lichess.org REST API**, with a **NestJS** (Spring Boot-style decorator framework) auth proxy implementing the **OAuth2 PKCE** flow. Diagnosed a missing feature in `chess.js`, opened [upstream PR #572](#), and published a fork on npm. **Portfolio site** operated as **Astro on GitHub Pages with GitHub Actions CI/CD**.
- **LLM code evaluation** for [DataAnnotation.tech](#) and [Stellar.ai](#) — prompt fixtures and grading rubrics across **.NET / C#, TypeScript, React, Node / NestJS, Python, Go, Rust, SQL, and MongoDB**, including **security audits with CWE / CVSS-graded inventories**. **60+ public PRs** to outside repos including the **Microsoft Authentication Library .NET** docs and `canada-ca/platform-security`.

Employment and Social Development, Canada Gatineau, QC

December 2021 – July 2025

Senior Software Developer — Cloud-Native Microservices (Benefits Delivery Modernization)

- Enhanced and maintained a **cloud-native microservices** enterprise onboarding application — **.NET Core / C# back end, Blazor front end, Azure SQL, Azure Functions, GraphQL** — serving procurement, HR, and IT-ops personnel. **Designed and consumed REST APIs between services** and **published and governed them through Azure API Management (API gateway)** with

authentication, throttling, and transformation policies; integrated enterprise IAM via **Azure AD / Entra (OAuth 2.0, OpenID Connect)**.

- **Serverless, event-driven workflows:** Blazor submissions kicked off **Azure Functions** triggered through **queue- and storage-based handoffs**, which created/updated Azure DevOps work items, ran pipelines, and provisioned users in on-prem Active Directory (via AAD Connect) by personnel profile — **bulk submissions fanning out asynchronously across the microservices**. Built a scheduled weekly work-item digest emailer driven by pipelines.
- **Containerized delivery and CI/CD:** operated end-to-end **Azure DevOps pipelines** for build, **Docker** containerization, **Kubernetes** deployment, and **Azure Data Factory** (ETL) orchestration. For the **AWS-hosted EI application**, deployed a **self-hosted pipeline agent into AWS** to run CI/CD there (completed AWS foundational training). Administered Azure DevOps for all BDM teams; advised on branching strategies and PR workflows; connected **Power BI to ADO Analytics**; ensured **WCAG / AODA** accessibility compliance.
- **Cloud security best practices:** migrated the application's downstream identity from rotating secrets (Key Vault + ADO PATs on 90-day rotation) to **Azure Managed Identities** — eliminating the secret-rotation incident class and properly attributing automated activity. Implemented **ITSG-33 (NIST-based) controls** — TLS / certificate management and **Key Vault** secret hygiene — for SA&A submissions against a **PBMM** profile.
- Designed and shipped **custom web extensions in React, TypeScript, and Node.js** — a build/deploy analytics widget rendering success-rate charts via the ADO Builds REST API, and a people-by-project reporter that replaced an unusably slow query with an in-memory hashmap. Ran weekly **demos and design walkthroughs**, mentoring junior developers.

Employment and Social Development, Canada Gatineau, QC August 2019 – November 2021

Senior Software Developer — Internal Web Application & CI/CD Automation

- Co-architected and developed an **internal C# / .NET web application** — web UI for authoring structured automation scenarios, **SQL Server** schema design, and a **co-designed REST API** providing bidirectional integration with **Azure DevOps pipelines** (versioned script-fetch endpoints and structured results-push endpoints).
- **Developed automated execution frameworks driven through CI/CD pipelines:** engineered Azure DevOps pipelines that provisioned Windows build agents to run **parallelized automated runs** against deployed applications and fed structured results back into work items. Performed **code review and standards enforcement** for the team.
- Drove a tooling consolidation that **cut software licensing costs** while preserving capability, and contributed to performance-optimization analysis of high-traffic deployed applications.

Department of Immigration, Refugees, and Citizenship, Canada

Ottawa, ON

August 2017 – August 2019

Team Lead / Technical Lead — Platform & Tooling

- **Technical leadership of a cross-functional team** — led **Agile / Scrum** sprint planning, review, retrospective, daily stand-up, and backlog refinement; introduced poker-planning estimation and capacity forecasting; set engineering standards and drove delivery cadence.
- Implemented and deployed **custom web extensions in React / TypeScript / Node.js** with responsive, accessibility-aligned UIs — leading the work hands-on while guiding the team.
- **Mentored developers in SQL Server query optimization** — query tuning, indexing strategy, partitioned tables, and execution-plan analysis against large databases and warehouses; coached on C# / .NET and SSRS.
- Administered departmental **Azure DevOps Server / TFS** instances and heavily customized process templates; managed user/group security and source-control access across multiple project collections.

Department of Justice, Canada Ottawa, ON

December 2015 – August 2017

Acting Lead Developer — iCase / Microsoft Dynamics CRM Modernization

- **Lead developer** on the live **Legal Case Management System (iCase)** — a large **n-tier ASP.NET / C# / ADO.NET** enterprise web application used by **5,000+ lawyers** — while concurrently building its **Microsoft Dynamics CRM** successor. Modified schema and **front-end UI to surface data-migration status in real time**, and adjusted integration stored procedures consumed by the department's SAP-based financial system.
- Engineered **Dynamics CRM plug-ins, custom actions, and workflows in C#** (rewriting legacy stored procedures as C# / LINQ); used **SSIS** for ETL between iCase and Dynamics; produced **entity-relationship diagrams** and **multi-month effort-estimation breakdowns** (FDD / TDD) and applied design patterns (Repository, Strategy, Mediator).
- **Established code-review processes gated on work items** — enforcing maintainability, security, and performance standards — and **mentored junior developers**.

Department of Citizenship and Immigration, Canada Ottawa, ON April 2014 – December 2015

Senior Software Engineer · Release Engineering Lead

- Hired alongside one other engineer to take over a release-management environment from a contractor team. **Decommissioned over-engineered legacy C# desktop apps** and replaced them with lightweight **PowerShell calling REST APIs** — most contractor tooling was simply deleted and **everything ended up in source control. Bias for action and ownership.**
- Implemented and deployed **web extensions in React / TypeScript / Node.js**; maintained a fork of Microsoft's `vsts-team-calendar` extension and contributed back upstream. Used **XSLT-via-PowerShell** to transform work-item-template XML for **automated deployment of custom process templates** — early build/deploy pipeline automation.
- **Installed, configured, customized, and upgraded** the Azure DevOps Server / TFS platform across dev / UAT / production — complex multi-step upgrades, **Kerberos / SPN configuration** on application tiers (coordinated with AD and DC admins without direct access), and database-tier migration.

Department of Justice, Canada Ottawa, ON October 2008 – April 2014

Full-Stack Developer — Legal Case Management System (iCase)

Core developer on the department's flagship **Legal Case Management System (iCase)** — a **distributed, multi-tier ASP.NET / C# enterprise web platform** used by **5,000+ government lawyers and contracted agent firms across Canada. High-availability and load-balanced:** deployed across **5 load-balanced web servers** and **11 regionally distributed document servers** nationwide, with role- and object-based security and integration to records-management, financial, and email systems, running **24/7**.

- Led design and largely solo implementation of the **"My iCase" personalization portal** — the new, **customizable widget-based entry point** to iCase. Ran JAD sessions, produced prototypes, authored BDD / TDD documentation, then built it on **ASP.NET Web Parts with a custom SQL personalization provider, Knockout.js MVVM** component binding (a direct ancestor of the **React** patterns I use today), and async AJAX web-service calls.
- **Optimized a high-traffic backend:** diagnosed a homepage report taking 10 s uncached — profiled with SQL Profiler, Report Execution logs, and execution plans, then replaced live joins with a nightly-populated flat table for a **50x speedup uncached (3-4x cached)**. Built a **self-initiated availability-monitoring suite** across the distributed web and document servers, and diagnosed and resolved a **COM Interop memory leak** affecting the document tier.
- **Early automation-framework development:** architected a **Java-based framework (IBM Rational Functional Tester)** that **parsed English-language scenario documents into executable scripts with round-trip code generation**, plus a companion manager in **C# / SQL Server**. Replaced a labour-intensive **two-week, 20-25-person manual pre-release validation cycle** with automated execution reviewed by one person in five days — **98% reduction in effort, \$160K saved per release cycle**, eliminating the standing code freeze.

Education

Certifications & Awards

Professional Scrum Master I (Scrum.org) · ITIL v3 Foundation (Loyalist Certification Services) · DOJ Team Merit Award 2012; Team Merit Award 2011; Team Spirit Award 2009

Technical Skills

Backend Frameworks: .NET Core / .NET 6 / 8, C#, ASP.NET Core, ADO.NET, Entity Framework, LINQ · Node.js, Express, NestJS · Spring Boot (familiarity)

Microservices & APIs: cloud-native microservices architecture, REST API design between services, GraphQL, WebSocket · **API gateway / management (Azure API Management)** — auth, throttling, transformation · event-driven & async messaging (queue- / storage-triggered) · load balancing across cloud & on-prem tiers

Containers & CI/CD: Docker, Kubernetes · **Azure DevOps Pipelines (YAML), GitHub Actions,** Jenkins (exposure) — build, automated test, containerized deploy · Azure Data Factory (ETL)

Cloud: Microsoft Azure — **Azure Functions (serverless),** App Service, Azure SQL, **API Management, Key Vault, Managed Identities, Entra / AAD,** Storage, Data Factory · **AWS** — self-hosted pipeline-agent deployment for the BDM EI application + foundational training · **GCP** — TWNR production hosting

Databases: SQL — MS SQL Server 2005 → 2019 (T-SQL, stored procedures, indexing, partitioning, execution-plan tuning), Azure SQL, PostgreSQL · **NoSQL — MongoDB** (working familiarity)

Security: OAuth 2.0 / OpenID Connect / PKCE, Entra / AAD IAM, Managed Identities, TLS / certificate management, Key Vault secret hygiene, ITSG-33 / PBMM controls, CWE / CVSS-graded security audits

Version Control: Git, GitHub, Azure DevOps Repos / TFVC · feature branches, PR-gated CI

Languages & Front-End: TypeScript, JavaScript, C#, SQL (T-SQL), PowerShell, Python, XML / XSLT, Go (light) · **React 19, Blazor,** Knockout.js, HTML / CSS, **responsive design, WCAG / AODA accessibility** · unit testing (xUnit / NUnit, Jest / Vitest)

Process / SDLC: Agile / Scrum (PSM I) — sprint planning, review, daily stand-up, backlog refinement · code review, mentoring, technical documentation, design walkthroughs · ITIL v3 Foundation